

بإمكانكم الاطلاع على تسجيلات المادة
من خلال منصة جو اكايمي

CHEMISTRY 101

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Question 1

of significant figures in (0.9001) = 4

Question 2

Ar gas has boiling point of 121 K, what is the temperature in C°?

$$K = C^{\circ} + 273 \dots\dots\dots C = -152 C^{\circ}$$

Q3: Choose the correct Answer:

- 1—Percentage Yield is Actual Yield multiplied by Theoretical yield divided by 100%
- 2—Empirical Formula give the actual whole number of mole ratio of atoms in molecule
- 3—Limiting Reactant, is the reactant consumed completely first limit and the amount of product
- 4—Molecular Formula give the relation whole the number of mole of atoms in molecules.

Correct Answer = # 3

Q4:

The mass percentage of (P) in $(\text{Ca}_3(\text{PO}_4)_2)$:

Molar mass of $\text{Ca}_3(\text{PO}_4)_2 = 3 * 40 + 31 * 2 + 16 * 8 = 310$

$$\% \text{ mass} = \frac{\# \text{ of P atoms} * \text{molar mass of P}}{\text{Molar mass of molecule}} * 100\%$$

$$\frac{2 \text{ atoms} * 31}{310} * 100\% = 20 \%$$

Q5:

A mass in g of 50 atoms of Fe:

$\# \text{ of atoms} = \# \text{ of mole} * \text{Avogadros number}$

$$50 = \# \text{ of mole} * 6.02 * 10^{23} \dots \dots \dots \# \text{ f mole} = 8.3056 * 10^{-23}$$

$$\# \text{ of mole} = \frac{\text{Mass}}{\text{molar mass}} \dots \dots \dots \text{Mass} = \text{mole} * \text{Molar Mass}$$

$$\text{Mass} = 8.3056 * 10^{-23} * 55.8 = 4.634 * 10^{-21} \text{g}$$

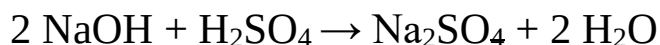
Q6:

What is empirical formula of oxalic acid, whose molecular formula $C_2H_2O_4$?

Empirical Formula = CHO_2

Q7:

How many cm^3 or ml of 0.7 M H_2SO_4 are needed to react completely with 25 cm^3 of 0.4 M NaOH according to this equation:



((1 L = 1000 ml = 1000 cm^3))

- Mole of NaOH = Molarity * Volume = $0.4 * 25 * 10^{-3} = 0.01$ mole
- 2 mole NaOH $\rightarrow$ 1 mole H_2SO_4
0.01 mole NaOH $\rightarrow$???
Of mole of $H_2SO_4 = 5 * 10^{-3}$ mole
- Volume of H_2SO_4 = mole / molarity
 $= 0.005 / 0.7 = 7.14 * 10^{-3}$ L
 $= 7.14 cm^3$

Q8:

A car travel 346.2 Km/6.27 hour, what is the maximum uncertainty in distance and time?

$$346.2 = 000.1 = 0.1$$

$$6.27 = 0.01$$

Q 9:

The molarity of 5.5 g of NaCl in 750 ml H₂O:

$$750 \text{ ml} = 0.75 \text{ L}$$

$$\text{Molar mass of NaCl} = 21 + 35 = 56 \text{ g/mole}$$

$$\text{Mole of NaCl} = \text{mass / molar mass} = 5.5 / 56 = 0.098 \text{ mole}$$

$$\text{Molarity} = \text{Mole / volume} = 0.098 / 0.75 = 0.13066 \text{ M}$$

Q 10:

Perform the following calculation, the number of significant figures:

$$(14.3 + 0.257 * 10^2) * 2.392 * 10^{-3}$$

$$(14.3 + 0.257 * 10^2) = 40 \text{ (1 sig.fig)}$$

$$40 * 2.392 * 10^{-3} = 95.68 * 10^{-3}$$

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$$9.568 * 10^{-2} = 9.57 * 10^{-2}$$

Q 11:

How many grams of sodium nitrate (NaNO_3) needed to prepare 250 cm^3 of $3 \times 10^{-3} \text{ M}$ solute:

$$250 \text{ cm}^3 = 0.25 \text{ L}$$

- Mole of NaNO_3 = Molarity * Volume
$$= 3 \times 10^{-3} \times 0.25 = 7.5 \times 10^{-4} \text{ mole}$$
- Molar Mass of NaNO_3 = $23 \times 1 + 14 \times 1 + 16 \times 3 = 85 \text{ g/mole}$
- Mass of NaNO_3 = Mole * Molar Mass
$$= 7.5 \times 10^{-4} \times 85 = 0.06375 \text{ g}$$

Q 12:

Calculate the mass in g of H in 30 g of ($\text{NH}_2\text{CH}_2\text{CO}_2\text{H}$):

- Molar mass of compound = $14 \times 1 + 12 \times 2 + 16 \times 2 + 5 \times 1 = 75 \text{ g/mole}$
- $$\text{Mass of H} = \frac{\# \text{ of H atom} \times \text{molar mass H}}{\text{molar mass of Compound}}$$
- $$\text{Mass of H} = \frac{5 \times 1}{75} = 0.066 \text{ g}$$

القيمة اللي طلعت معنا هي ل 1 غرام بس من المركب ... احنا عنا 30 غرام من المركب

- 1 g Compound $\rightarrow$ 0.066 g H
30 g compound $\rightarrow$?? G H

1.98 g of H in 30 g compound

Q 13:

Choose the incorrect answer:

- Each element has its own spectrum
- Photon is the smallest increment energy gained or lost from object
- Each bond of spectrum has own wavelength
- Energy isn't Quantized

Answer = # 4

Q 14:

Red Laser pointer emits light with wavelength = 550 nm, the frequency of this light?

$$c = \lambda V \dots \dots V = \frac{c}{\lambda}$$

$$V = \frac{3 * 10^8}{550 * 10^{-9}} = 5.45 * 10^{14} \text{ Hz}$$

Q 15:

The energy (J) absorbed for the electronic transition in H atom from (n= 3 to n = 4):

$$E = -2.18 * 10^{-18} \left(\frac{1}{nf^2} - \frac{1}{ni^2} \right)$$

$$E = -2.18 * 10^{-18} \left(\frac{1}{(4)^2} - \frac{1}{(3)^2} \right) = 1.0573 * 10^{-19} \text{ J}$$